

Question:

Assume that you have four number display cards, each card having a number written on it. You cannot view more than two cards at a time. How many comparisons would you require to arrange the cards in ascending order using this constraint?

Remember: You can only view and compare two cards at a time. The initial card values are:

40, 30, 20, 10

Type of Sorting Used: Selection Sort

Why It's Selection Sort:

- **Selection sort** works by **repeatedly finding the minimum element** from the unsorted part and putting it at the beginning.
- In your example:
 - You start by comparing two cards at a time.
 - From the full list [40, 30, 20, 10], you **repeatedly select the smallest number** by comparing two elements at a time.
 - After selecting the smallest (10), you **repeat the process for the remaining elements** [40, 30, 20], then [40, 30], and so on.

Step-by-step Explanation:

Initial List:

[40, 30, 20, 10]

Pass 1 – Find the smallest element:

- Compare 40 and 30 → smaller is 30
- Compare 30 and 20 → smaller is 20
- Compare 20 and 10 → smaller is 10
- ☒ **10 is the smallest**, put it at position 0.

Now: [10, 40, 30, 20]

Pass 2 – Find the next smallest from remaining [40, 30, 20]:

- Compare 40 and 30 → 30
- Compare 30 and 20 → 20
- ☒ **20 is the next smallest**, put it at position 1.

Now: [10, 20, 40, 30]

Pass 3 – Find next smallest from [40, 30]:

- Compare 40 and 30 → 30
- ☒ Put 30 at position 2.

Now: [10, 20, 30, 40]

Pass 4 – Only one element left (40), already sorted.

Key Points:

- It mimics **Selection Sort**.
- You use **only 2 cards at a time** (pairwise comparison).
- Total comparisons = $n(n-1)/2 = 4(4-1)/2 = 6$ comparisons.